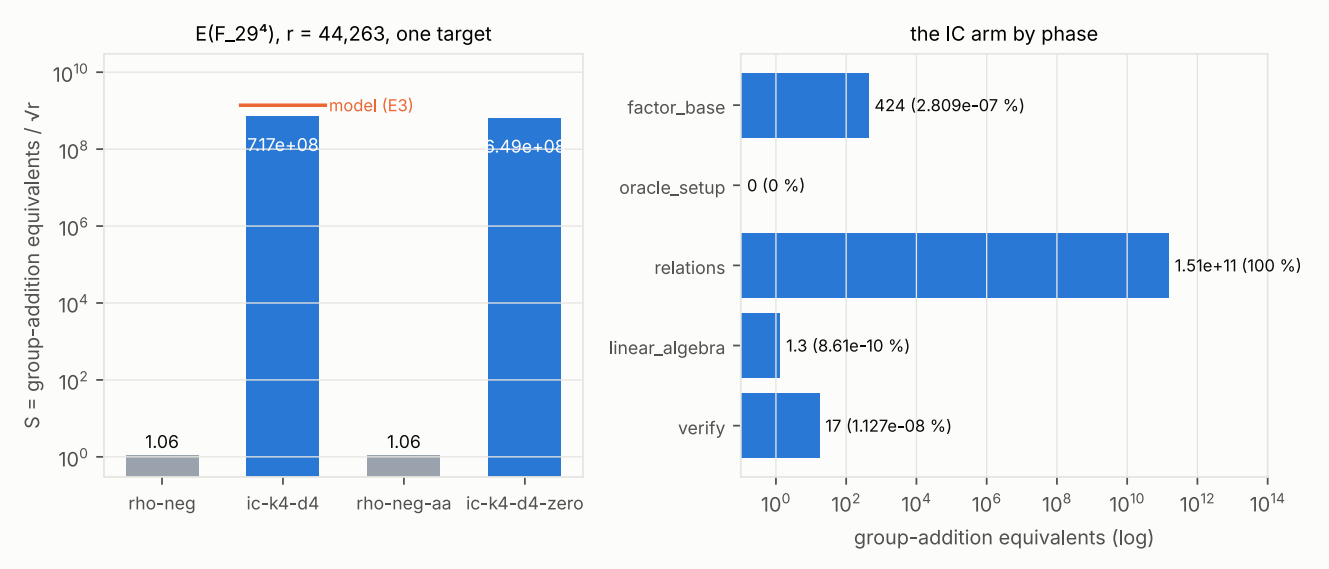
Gaudry's $k = 4$ index calculus with the D_4 oracle, end to end in ecbench

2026-10-13. `research/gaudry_quartic_d4_ecbench_20261013/`. Protocol registered before the session; audit OK (3 of 3 runs replayed identically). **Class: engineering.**

Setup

$E(\mathbb{F}_{29^4})$ with rational 2-torsion, $\#E = 16 \cdot 44263$ (registered `icv1-fp5k4-tm926-7bc5b359`), one target. `ic.pipeline` with FGHR's Y-line base (columns $\{\pm P, \pm(P+T)\}$) and the `gaudry-d4` oracle (S_5 in the D_4 invariants, Macaulay degree 14); every phase charged, the oracle's $\mathbb{F}_p$ multiplications at the curve's 97 per addition. Reference: rho with the negation map.



Left: S per arm, with §11.16′s model value for the IC arm. Right: the IC arm by phase.

arm	method	S	isolation
rho-neg	`rho.negation`	1.065	L2
ic-k4-d4	`ic.pipeline`, `fghr-y-line`, `gaudry-d4`	$\mathbf{7.17 \cdot 10^8}$	L1
rho-neg-aa	`rho.negation`	1.065	L2

633 relation trials, 631 solves at $2.32 \cdot 10^{10}$ $\mathbb{F}_p$ multiplications each, 8 relations, rank 8 on 8 unknowns; relations are $\approx 100\%$ of S. **Index calculus costs $\mathbf{6.7 \cdot 10^8}$ × rho**, within 2× of the model's $1.3 \cdot 10^9$; every registered range held, though the decomposition rate the trial range assumed (0.003–0.008) was low against the 0.017 seen. S omits the 9 degenerate solves (under 1.4 %). Three decompositions were lost to a summand outside the base (probably $Y = 0$); 9 of 631 systems were degenerate.

Amendment 1. With the order-4 point $Y = 0$ in the base (`fghr-y-line:zero=1`): 574 trials, no lost decomposition, $S = 6.49 \cdot 10^8$ (−9.5 %), audit OK. On 24,000 enumerated targets the point raises the decomposition rate 1.42 ×.

Scope: one small curve and one target; the method working and priced end to end where it was expected to lose by nine orders of magnitude. It confirms the model's relation-phase constant, not the projected crossover near 2^{135}.

Environment: Claude Code cloud container, Xeon @ 2.10 GHz, one reserved core, rustc 1.97.0. Conductor unreachable: no task card.